

Science and Complexity

SENSS Specialist Training: Introduction to Computational Methods

19 May 2026

Baris Ari

Part 2

Which field is easier to deal?
Physics or Sociology?



Neil deGrasse Tyson 

@neiltyson



In science, when human behavior enters the equation, things go nonlinear. That's why Physics is easy and Sociology is hard.

12:03 AM · Feb 6, 2016



531



10K



17K



65



Social Sciences

- Social Sciences are not as successful as other branches of Science (Physics, Chemistry)
- Not very good at producing reliable results
- **Warren Weaver:** [Social] science can handle simple and large problems, but not realistic ones (ones that really matter)

Weaver W. *Science and Complexity*. American Scientist. 1948

SCIENCE AND COMPLEXITY

By WARREN WEAVER

Rockefeller Foundation, New York City

SCIENCE has led to a multitude of results that affect men's lives. Some of these results are embodied in mere conveniences of a relatively trivial sort. Many of them, based on science and developed through technology, are essential to the machinery of modern life. Many other results, especially those associated with the biological and medical sciences, are of unquestioned benefit and comfort. Certain aspects of science have profoundly influenced men's ideas and even their ideals. Still other aspects of science are thoroughly awesome.

How can we get a view of the function that science should have in the developing future of man? How can we appreciate what science really is and, equally important, what science is not? It is, of course, possible to discuss the nature of science in general philosophical terms. For some purposes such a discussion is important and necessary, but for the present a more direct approach is desirable. Let us, as a very realistic politician used to say, let us look at the record. Neglecting the older history of science, we shall go back only three and a half centuries and take a broad view that tries to see the main features, and omits minor details. Let us begin with the physical sciences, rather than the biological, for the place of the life sciences in the descriptive scheme will gradually become evident.

Problems of Simplicity

Speaking roughly, it may be said that the seventeenth, eighteenth, and nineteenth centuries formed the period in which physical science learned variables, which brought us the telephone and the radio, the automobile and the airplane, the phonograph and the moving pictures, the turbine and the Diesel engine, and the modern hydroelectric power plant.

The concurrent progress in biology and medicine was also impressive, but that was of a different character. The significant problems of living organisms are seldom those in which one can rigidly maintain constant all but two variables. Living things are more likely to present situations in which a half-dozen, or even several dozen quantities are all varying simultaneously, and in subtly interconnected ways. Often they present situations in which the essentially important quantities are either non-quantitative, or have at any rate eluded identification or measurement up to the moment. Thus biological and medical problems often involve the consideration of a most complexly organized whole. It is not surprising that up to 1900 the life sciences were largely concerned with the necessary preliminary stages in the application of the scientific method—preliminary stages which chiefly involve collection, description, classification, and the observation of concurrent and apparently correlated

Based upon material presented in Chapter I, "The Scientists Speak," Boni & Gaer, Inc., 1947. All rights reserved.

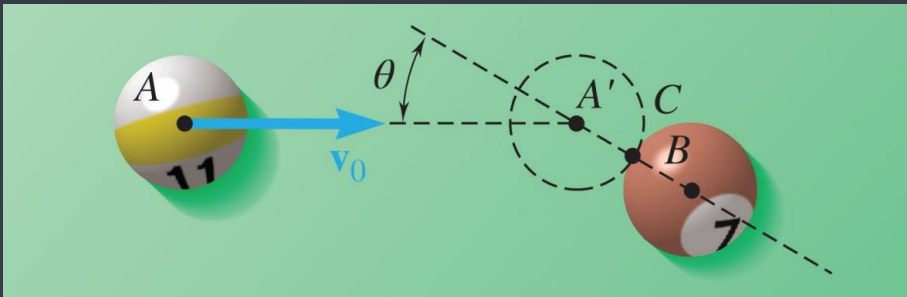
Warren Weaver on History of Science

1. Problems of simplicity: two or a few variable problems
2. Problems of disorganized complexity: a very large number of variables (infinitely many)
3. Problems of organized complexity: middle-range

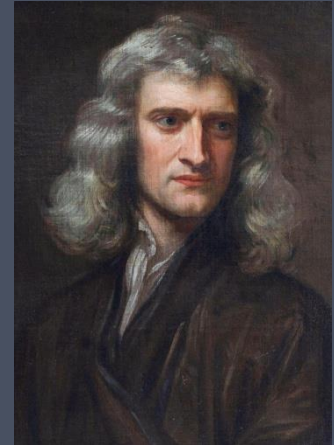
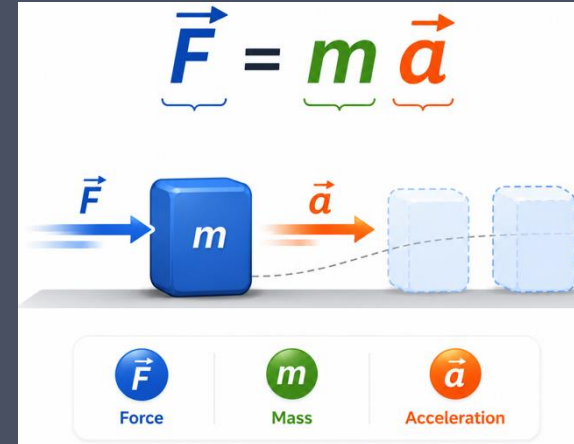
1. Problems of simplicity

Two or a few variable problems:

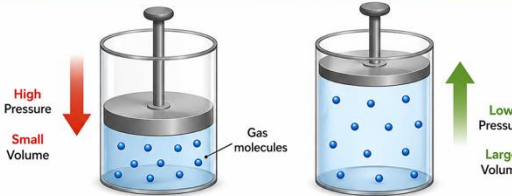
- reduce the problem to two variables and explain
- Solvable analytically (closed form equations)
- Linear, well-behaved interactions
- The triumphant achievements of 17th-19th century science



The motion of two billiard ball on a frictionless table



BOYLE'S LAW



High Pressure
Small Volume

Low Pressure
Large Volume

Gas molecules

When you push the piston down, pressure **increases** and volume **decreases**.

When you pull the piston up, pressure **decreases** and volume **increases**.

BOYLE'S LAW

At constant temperature (and amount of gas), the volume of a gas is **inversely proportional** to its pressure.

$$V \propto \frac{1}{P}$$
$$V = \frac{k}{P}$$

where k is a constant (for a given gas at constant temperature).

2. Disorganized Complexity

Disorganized complexity refers to a problem in which:

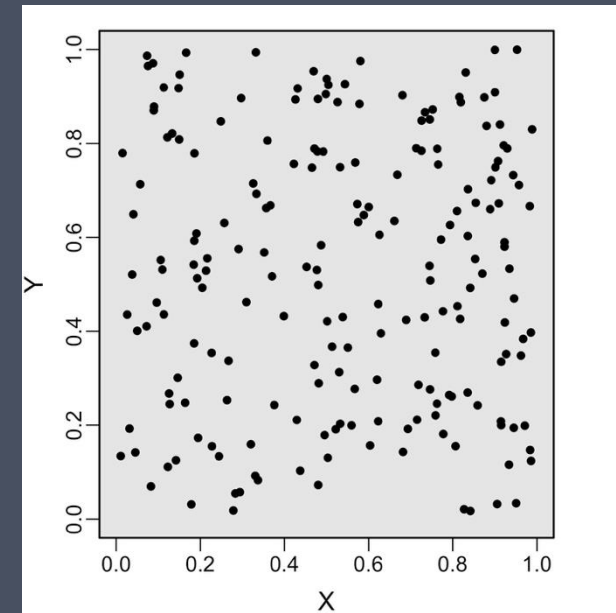
- the number of variables is very large (*thousands to billions; infinitely many*)
- each of the many variables has a behavior which is individually erratic, or perhaps totally unknown

However, in spite of this helter-skelter, or unknown, behavior of all the individual variables:

- the system, as a whole, possesses certain orderly and analyzable *average* properties.

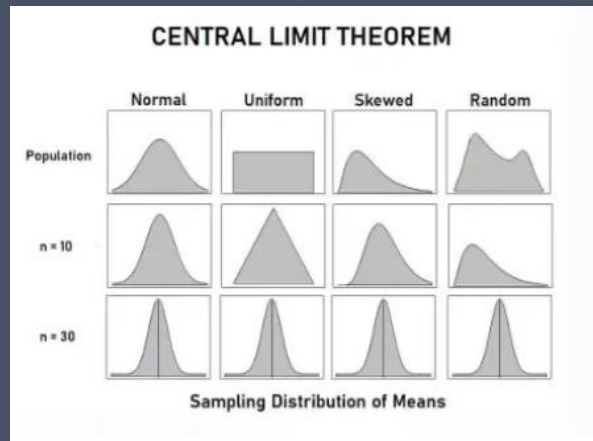
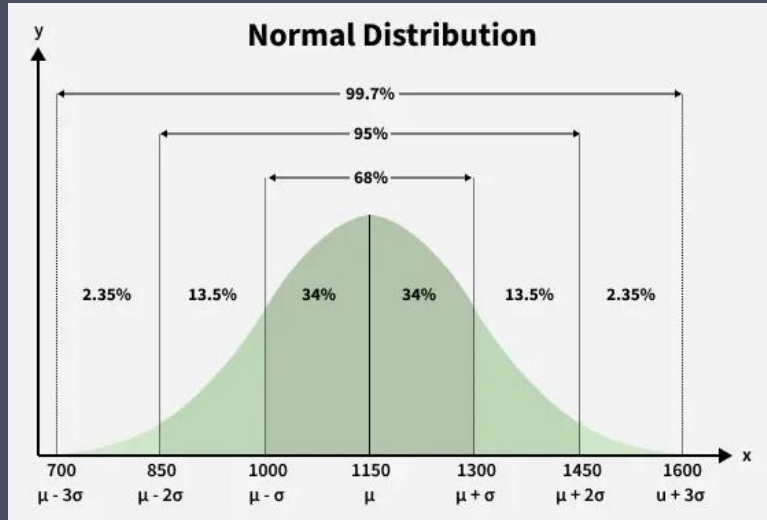
Note: variables here refers to the number of observations (draws / realisations of a random variable) rather than the number of predictor variables in a regular regression analysis.

The motion of 10^{23} billiard balls on a frictionless table



*We cannot predict each and every ball's precise location
We can predict the average number of balls in a given area*

2. Disorganized Complexity



Modern statistics can cope with disorganized complexity

- 20th century onwards: Probability theory
- **Probability theory**: normal distribution, central limit theorem, etc.

Science of averages:

- Describing average behaviour of a system (e.g., regression)
- Focus on averages & long-run characteristics
- Individual anomalies cancel each other out

Examples:

- motion of atoms or gas molecules
- insurance claims
- some market behaviour: average ticket sales

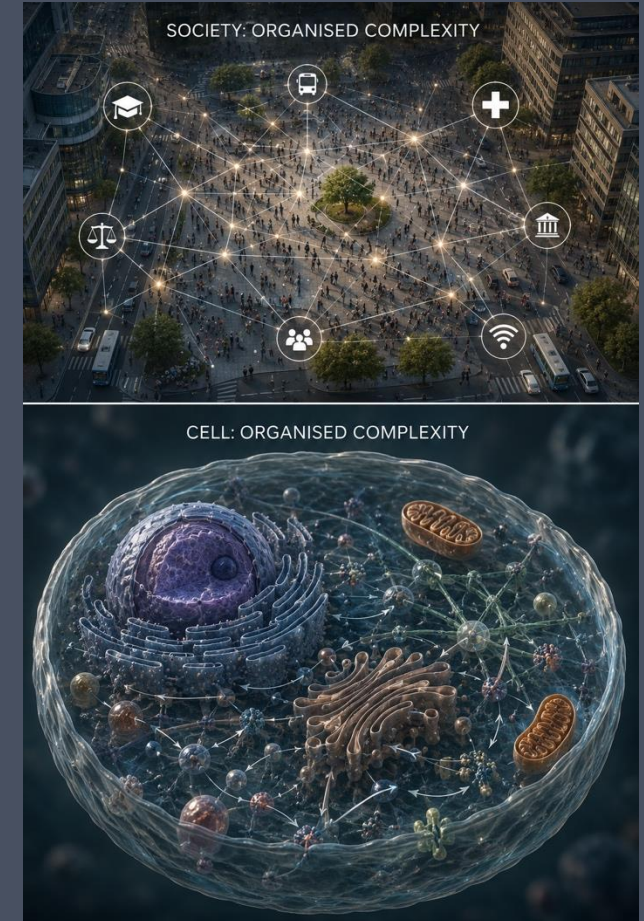
2. Disorganized Complexity: Prediction without Causality

[S]tatistical techniques are not restricted to situations where the scientific theory of the individual events is very well known, as in the billiard example where there is a beautifully precise theory for the impact of one ball on another.

[They] can also be applied to situations, like the insurance example, where the individual event is as shrouded in mystery as is the chain of complicated and unpredictable events associated with the accidental death of a healthy man.

3. Problems of Organized Complexity

- Middle-range
 - Still large number of variables but not infinitely many
 - **Key concept:** variables are interdependent
- **Organized complexity:** *problems that involve dealing simultaneously with a sizable number of factors which are interrelated into an organic whole*
- *Shows the essential feature of an organization:*
 - Biological organisms
 - Societies



3. Problems of Organized Complexity

Hallmarks of Organized Complexity:

- System formed by organized and interacting parts
- Interactions and interconnectedness
- Feedback
- Non-linearity
- Heterogeneity
- Dynamic system (not static)
- Reducing to averages → not very useful
- Cannot solve in closed form
 - hard or intractable

Examples:

- War- and peace-making
- Climate systems
- Racial discrimination
- Bank runs (market behaviour during crisis)
- Market prices (e.g., bitcoin, gold, etc.)
- Ant colonies
- A biological organism
- Immune system response
- Disasters (flooding; fire; earthquake)

Third Great Advance

- Science has achieved two great leaps
 1. Two (or, a few) variable problems
 2. Infinitely many variable problems
- We need a third leap to tackle complex problems
 - Interdependency
- *Complex problems*: those that really matter for biological and social systems

Third Great Advance

! These problems—and a wide range of similar problems in the biological, medical, psychological, economic, and political sciences—are just too complicated to yield to the old nineteenth-century techniques which were so dramatically successful on two-, three-, or four-variable problems of simplicity. These new problems, moreover, cannot be handled with the statistical techniques so effective in describing average behavior in problems of disorganized complexity.

{ These new problems, and the future of the world depends on many of them, requires science to make a third great advance, an advance that must be even greater than the nineteenth-century conquest of problems of simplicity or the twentieth-century victory over problems of disorganized complexity. Science must, over the next 50 years, learn to deal with these problems of organized complexity.

How to tackle organized complexity?

- Computational approaches!
- Promising development: *new types of electronic computing devices*

These devices are, in flexibility and capacity, more like a human brain than like the traditional mechanical computing device of the past.

They have memories in which vast amounts of information can be stored. They can be "told" to carry out computations of very intricate complexity, and can be left unattended while they go forward automatically with their task.